Continuous Autoregressive Language Models (CALM) vs. HUFD Constant-Attention Architecture

Abstract

Continuous Autoregressive Language Models (CALM) redefine large-language-model generation by predicting next vectors rather than next tokens, improving efficiency through continuous latent manifolds. The HUFD architecture, built around Constant Attention, Self-Organizing Memory (SOM), and RAGU—pursues the complementary goal of holonomic coherence, interpretability, and forensic auditability in cognitive systems. This report contrasts CALM's likelihood-free, energy-based inference with HUFD's lucidity-modulated holonomic framework, highlighting how CALM cuts financial burn per output; whereas HUFD cuts structural burn per operation (greater savings).

1. Paradigm Alignment

CALM (Shao et al., 2025) recasts language modeling from discrete next-token prediction to next-vector prediction in a continuous latent space. Each vector encodes K tokens via a variational autoencoder, enabling "continuous autoregression." Training is likelihood-free, relying on energy-based scoring rules instead of cross-entropy. Its claim is that vector prediction increases "semantic bandwidth" per generation step, thus reducing sequence length and FLOPs by ≈ K×.

HUFD (Constant Attention + SOM + RAGU) instead treats cognition and sensing as a holonomic manifold: every representational step is a closed-loop attention cycle with embedded provenance. Rather than compressing tokens, HUFD compresses decision context; SOM projects the manifold into archetypal basins ("Hero," "Shadow," "Mentor"), and RAGU writes semantic updates along those coordinates. Where CALM seeks computational throughput, HUFD seeks interpretability, traceability, and field stability, a <u>completely different optimization objective</u>.

2. Architectural Contrast

CALM's stack—autoencoder → energy-transformer → Brier scoring—corresponds roughly to HUFD's Constant Attention → SOM → RAGU pipeline, but *with inverse priorities*. CALM compresses symbol streams for efficiency (Razavi et al., 2019; Child et al., 2019); HUFD compresses decision context to maintain narrative stability (Hinton 2021; Ritter et al., 2021).

SOM in HUFD behaves analogously to CALM's latent manifold but introduces archetypal attractors (cf. Kohonen 1990; Jung 1959) that act as fixed-point centers of human-interpretable meaning. RAGU serves as an energy transformer constrained by lucidity feedback, preventing representational drift.

3. Mathematical & Epistemic Differences

CALM relies on continuous energy distances between latent vectors as implicit likelihoods, aligning with energy-based learning (LeCun et al., 2006). HUFD instead computes *Lucidity*, a scalar combining SOM quantization error and archetype alignment, used to gate RAGU's learning rate.

Conceptually, CALM minimizes energy discrepancy in latent space; HUFD maximizes coherence entropy over archetypal manifolds - a cognitive analog of minimizing free energy (Friston 2010). Both replace discrete symbolic losses with geometric alignment objectives, but <u>HUFD enforces interpretability as a conservation law.</u>

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4. Comparative Advantages

Category	CALM	HUFD
Throughput	K-fold speed-up per step (Shao et al., 2025)	Deterministic attention trace
Robustness	Latent continuity via VAE regularization	Lucidity feedback suppresses collapse
Interpretability	Moderate (autoencoder visualization)	High (auditable archetype trajectories)
Energy Efficiency	Reduced FLOPs (≈ 30–50 %)	Linear update complexity
Application Focus	LLM compression, continuous inference	Trusted autonomy, AI governance, continuous training

4. CALM's Cost Model

Mechanism: CALM collapses K discrete tokens into one latent vector. If K = 8–16, the model runs 8–16× fewer forward passes per generated sequence. Because it uses the same Transformer backbone (with an added autoencoder head), GPU memory and parameter counts stay roughly constant, but FLOPs and wall-time drop by about 50–70 % for comparable semantic output length. Result: In inference mode on a 7 B-parameter baseline, CALM shows:

i) FLOP reduction: ≈ 60 % (Shao et al., 2025)
ii) End-to-end inference cost: ≈ 40–55 % lower than a discrete-token LLM when amortized over context compression
iii) Training cost: higher per-step (+15–25 %) due to continuous VAE pre-training, but converges in ~½ the steps → net 20–30 % cheaper full-cycle training

CALM's economics: roughly ½ compute per tokenized output, ⅔ the total cloud bill once you deploy at scale. It wins on throughput and GPU utilization, not on energy-consumed per operation.

5. Constant-Attention HUFD Cost Model

HUFD is architecturally leaner but philosophically heavier. It doesn't compress tokens; it stabilizes state. The Constant-Attention core replaces multi-head softmax attention with a single normalized vector-field update ($O(n)$ rather than $O(n^2)$ in sequence length). RAGU introduces a lightweight GRU-class recurrence (≈ 15% of Transformer cost per layer) and SOM adds a sparse projection update (≈ 5%).

Empirical scaling (internal HUFD prototypes, 32-D embeddings):

i) Training cost: ≈ 60 – 70% lower than an equivalent Transformer because attention is linear-time and no gradient checkpointing needed.
ii) Inference cost: ≈ 55 – 65% lower for constant-time recurrent steps (bounded memory, no key/value cache growth).
iii) Energy cost: per-token joules drop ≈ 65 – 70% versus GPT-class baseline; gains come from cache locality and deterministic updates.
iv) Storage/audit overhead: adds ~8–12 % (audit traces + Merkle hashes).

The net HUFD compute bill is ≈ ⅓ of conventional Transformer, roughly par with CALM in raw FLOPs, but the spend shifts, CALM's cost savings are in sequence compression; HUFD's savings are in algorithmic complexity and persistent-memory efficiency.

6. Economic Comparison Table

Metric	Baseline Transformer	CALM	HUFD
FLOP reduction	—	≈ 60 %	≈ 65 %
Wall-time per 1k tokens	1×	0.45×	0.40×
Training epochs to convergence	1×	0.7×	0.6×
Peak GPU memory	1×	0.95×	0.6×
Energy / token (J)	1×	0.55×	0.30×
Audit / explainability cost	none	low	+10 % overhead
Overall cloud-inference cost	100%	≈ 65 %	≈ 35–45 %
CapEx leverage (hardware reuse)	baseline	moderate	high - CPU-friendly, FPGA-viable

7. Venture & Program-Officer Relevance

From a portfolio standpoint, CALM is a horizontal scaling improvement on prevailing transformer/ Atention (akin to Efficient-Transformer research - Tay et al., 2020, refining a conventional Transformer architecture incrementally), while HUFD is a real-time continuous-training, small-carbon-footprint, vertical assurance layer compatible with explainable-AI initiatives (Doshi-Velez & Kim 2017).

Government and VC entities (DARPA XAI 2023; In-Q-Tel 2024) now value continuous verification over raw scale; HUFD's Constant-Attention architecture directly supports that shift by providing cryptographically anchored provenance and self-auditing loops.

CALM addresses compute cost; HUFD addresses costs + trust deficit, the next capital moat for AI.

i) CALM = throughput efficiency. It halves GPU hours for language inference but still depends on transformer-class accelerators.

ii) HUFD = CapEx leverage & structural efficiency by eliminating $O(n^2)$ scaling of CALM and introduces reusable, audit-grade recurrence - cost savings continue to grow as sequence length or agent count rises.

iii) Crossover point: at context lengths > 1,000 tokens or multi-agent deployments, HUFD's linear-time 'Constant Attention' dominates CALM's vector compression on both cost and latency.

iv) Additional upside: HUFD logs provenance without an extra model pass, satisfying explainability mandates that CALM would need a second audit model to replicate which gives HUFD a 15–25 % compliance cost advantage in regulated deployments.

CALM cuts financial burn per output; HUFD cuts structural burn per operation (greater savings). At equal optimization maturity, "Constant Attention" delivers ~1.5–2× better total cost of ownership than CALM in long-context or real-time applications, while adding the governance and audit layer investors are now willing to pay a premium for.

8. Integrative Outlook

A hybrid model could potentially fuse CALM's continuous latent prediction with HUFD's Lucidity-modulated gating: predict next vector only when self-coherence exceeds threshold. Such coupling merges CALM's conventional Transformer/Attention efficiency with HUFD's epistemic safety, aligning with emerging "Consciousness-in-the-Loop" research (Dehaene et al., 2024) and neuro-symbolic holonomy frameworks. However, CALM benefits more from integrating HUFD than HUFD gains via CALM.

Result: Continuous Cognitive Model (CCM) combining vector efficiency and interpretive accountability.

9. Summary for Review Board

Criterion	CALM	HUFD
Technical Readiness	~TRL 4	~TRL 4
Novelty	Continuous autoregression	Continuous real-time training, Holonomic attention with Lucidity control
Program Fit	LLM optimization	Trusted autonomy, explainable ISR
Commercial Bridge	Cloud-inference efficiency	AI-governance for classified and commercial AI deployments.
Strategic Leverage	Cost efficiency	Cost efficiency, cognitive assurance & auditability

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